

Open Challenge Navigation Algorithm

Overview

The robot is designed to complete three autonomous laps around the WRO Future Engineers track.

The navigation system combines:

- Three HC-SR04 ultrasonic sensors
 - MPU6050 gyroscope
 - MG90S steering servo
 - Two DC drive motors
 - Arduino Uno
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Sensor Configuration

Front Ultrasonic Sensor

Used to detect:

- Walls
- Corners
- Obstacles directly ahead

Left Ultrasonic Sensor

Used to measure the distance to the left wall.

Right Ultrasonic Sensor

Used to measure the distance to the right wall.

MPU6050

Used to:

- Measure heading changes
 - Verify 90° turns
 - Count completed corners
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Steering Calibration

Position	Servo Value
Maximum Right	0
Straight	80
Maximum Left	150

Main Navigation Strategy

1. Start

The robot initializes all sensors and centers the steering system.

2. Wall Following

The vehicle continuously reads the right ultrasonic sensor.

If the robot gets too close to the wall:

- Steering is adjusted slightly left.

If the robot moves too far from the wall:

- Steering is adjusted slightly right.

If the distance is within the target range:

- The robot continues straight.

3. Corner Detection

A corner is detected when:

- The right wall disappears.
- The front sensor detects an approaching wall.

4. Turning

When a corner is detected:

1. Steering moves to maximum right.
2. The robot continues moving.
3. The MPU6050 measures the heading change.
4. When approximately 90 degrees are reached:
 - Steering returns to center.

- Corner counter increases by one.

5. Lap Counting

Each completed corner increments the turn counter.

- 4 turns = 1 lap
- 8 turns = 2 laps
- 12 turns = 3 laps

6. Finish

After completing 12 turns:

- Motors stop.
- Steering returns to center.
- The robot waits for reset.

Pseudocode

Initialize Sensors

while(turns < 12)

 Read Front Sensor
 Read Left Sensor
 Read Right Sensor

 if CornerDetected()

 TurnRight90Degrees()

 turns++

 else

 FollowRightWall()

Stop Motors